

TrES-3b

R-2555

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110414 · created 2026-08-26T01:30:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455665.81908 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.022
Transit depth (Rp/Rs)^2	2.32 +/- 0.66 [%]
Orbital Inclination (inc)	82.36 +/- 1.03
Airmass coefficient 1 (a1)	1.029 +/- 0.013
Airmass coefficient 2 (a2)	-0.0155 +/- 0.005
Scatter in the residuals of the lightcurve fit is	1.24 %
Best Comparison Star	#2 - [459, 406]
Optimal Aperture	3.74
Optimal Annulus	9.0
Transit Duration (day)	0.057 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.022 vs 0.16309 (deviation 0.5σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.03σ)
Mid-time O–C	-2.3 min
Depth SNR	6.9
Residual scatter	1.24 %

Light curve

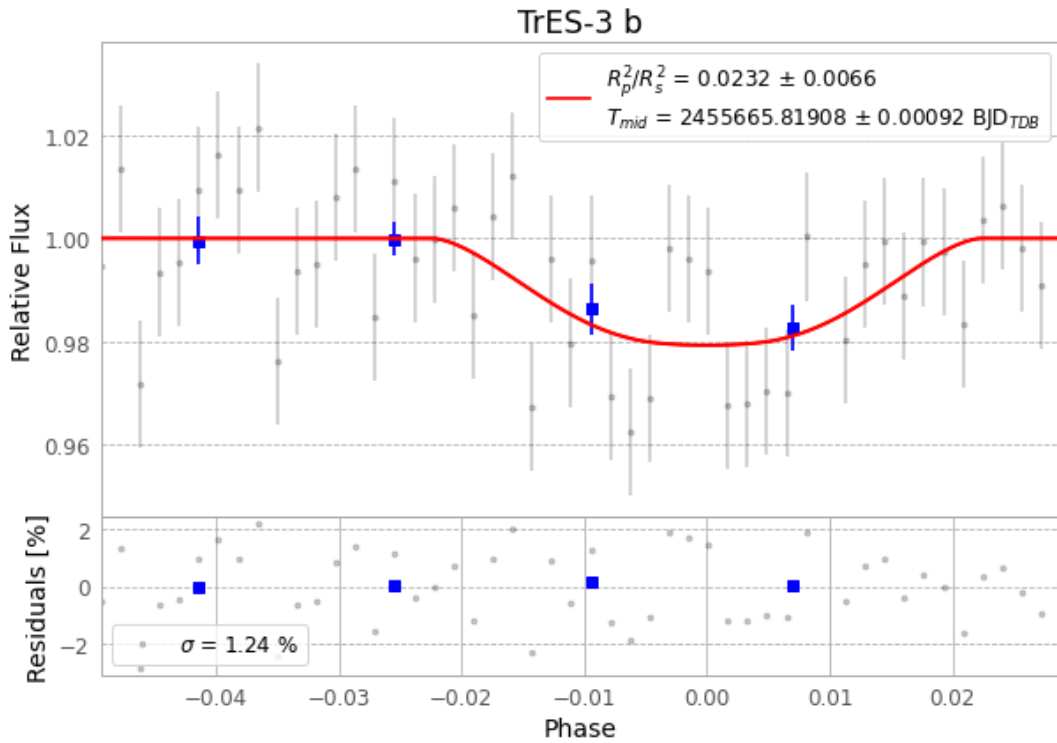


Figure 1 — FinalLightCurve_TrES-3 b_2011-04-13.png (SHA-256 3241adf0c10d1cd0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [222, 186], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b26bdb3fbd7b2b5f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), TRES-3110414060313.FITS → TRES-3110414083013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110414.log (659be0930476...), FinalLightCurve_TrES-3 b_2011-04-13.png (3241adf0c10d...), FinalLightCurve_TrES-3 b_2011-04-13.csv (9da3f7483aee...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 01:30Z.